

TETHYS SEA FRAGMENTATION AND SPECIATION OF *PANULIRUS* SPINY LOBSTERS

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ABSTRACT

A composite of the phylogenetic relationships of the species of *Panulirus*, based on morphological and genetic comparisons is presented. Data on the timing of the longitudinal break up of the Tethys Sea into the separate major oceans (and the resulting ecosystem changes) allow a review of the isolating mechanisms that may have resulted in the present day species distribution. These scenarios have been supplemented by personal visits to most of the regions inhabited by the various species of *Panulirus*. Two major phylogenetic lineages are recognized, the first emerging earlier and exhibiting a habitat preference for clear waters associated with continents or volcanic islands, the second major lineage emerging a little later and showing a distinct preference for the turbid waters of coastal habitats. The genus originated in the broad Tethys Sea and the *Panulirus* stock was initially divided longitudinally by the collision of Africa with Eurasia, isolating the Indo-west Pacific element; the final separation of the east Pacific from the Atlantic elements occurred with the Pliocene closure of the Panama Seaway. New habitats for incipient species were formed by these geological events as well as by the emergence of oceanic islands, climatic changes, and the formation of more discrete current systems.

RÉSUMÉ

Les inter-relations phylogénétiques des espèces de *Panulirus*, basées sur des comparaisons morphologiques et génétiques sont présentées. Les données sur le moment du clivage longitudinal de la mer Téthys en deux grands océans séparés (et les conséquences sur l'écosystème) permettent une revue des mécanismes d'isolation à l'origine de la distribution des espèces actuelles. Ces scénarii ont été confortés par des visites personnelles dans la plupart des régions colonisées par les différentes espèces de *Panulirus*. Deux grandes lignées phylogénétiques sont reconnues, la première émerge assez tôt et montre une préférence pour les habitats d'eau claire associés à des continents ou des îles volcaniques, la deuxième lignée émerge un peu plus tard et montre une préférence pour les eaux turbides des habitats côtiers. Le genre originait dans la mer Téthys encore entière et le stock de *Panulirus* a été divisé longitudinalement par la collision de l'Afrique avec l'Eurasie, isolant l'élément Indo-ouest Pacifique; la séparation finale entre l'est Pacifique et les éléments Atlantiques

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s'est produite avec la fermeture du détroit de Panama au Pliocène. De nouveaux habitats pour les nouvelles espèces ont été formés par ces événements géologiques et par l'émergence des îles océaniques, les changements climatiques et la formation de systèmes de courants plus discrets.

INTRODUCTION

The morphological and genetic comparisons by Baisre (1994), McWilliam (1995), Ptacek et al. (2001), and Patek & Oakley (2003) have enabled a review of the phylogenetic relationships in *Panulirus* that is acknowledged as the most recently evolved palinurid genus (George & Main, 1967; Pollock, 1992; Baisre, 1994; McWilliam, 1995). Following their genetic comparisons, Ptacek et al. (2001) recognized two major lineages within *Panulirus* and suggested that the Atlantic *P. argus* (Latreille, 1804) was the most likely species to represent the ancestral stock.

George & Main (1967) had proposed for *Panulirus*, an origin in the Indo-west Pacific and a four-step radiation resulting from post-Pliocene climatic changes with the continental landmasses fixed in their present positions. After taking into account continental movements and contemporary taxonomic and genetic assessments, George (1997) extended the probable emergence time of *Panulirus* to at least the late Miocene rather than in the Pleistocene as suggested by George & Main (1967), Pollock (1992), and Sekiguchi (1995). In 1995, visits were made to the east Pacific and the Atlantic to gather ecological comparative data on the *Panulirus* species of those regions and in 1998 I visited the Ogasawara Islands to study the ecology and fishery of a new species, *P. brunneiflagellum* Sekiguchi & George, 2005.

The primary analyses of kinship and ancestor-descendant relationship are based on classical (non-cladistic) methodology that has been supplemented with personal experience of the biology, ecology, and fisheries practices of most of the *Panulirus* species. This review takes particular note of the importance of the longitudinal Tertiary fragmentation of the Atlantic-European region of the Tethys Sea (fig. 1) and interprets the palaeogeomorphic and palaeo-oceanographic events that likely led to the speciation and radiation of 27 taxa of *Panulirus*.

RESULTS AND DISCUSSION

Panulirus taxonomy and widespread variable species

Most taxa of *Panulirus* are panmictic, have clear allopatric distributions, and are self-recruiting via local larval transport systems. Table I lists the currently recognized taxa of *Panulirus*, together with their distribution, local transport

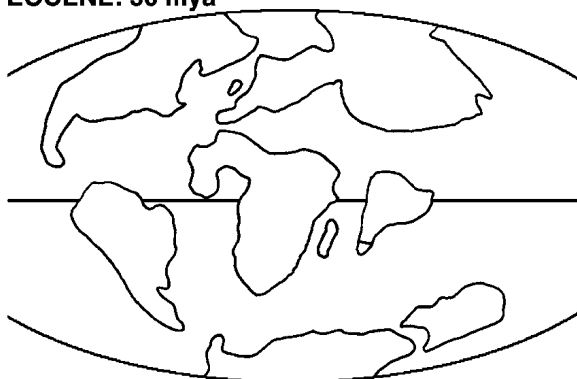
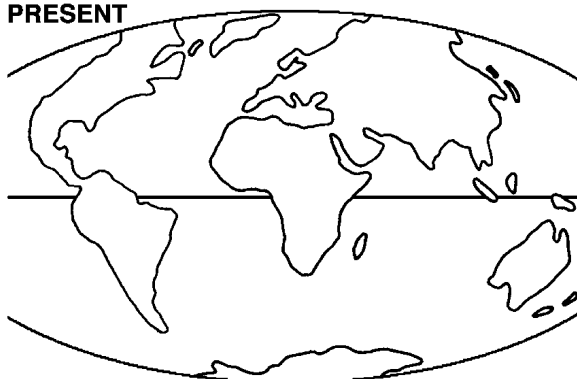
EOCENE: 50 mya**MIOCENE: 20 mya****PRESENT**

Fig. 1. Contraction and fragmentation of the Tethys Sea caused by continental movements from Eocene to Present, after Evans (1987).

TABLE I

Main distributions, larval transport system, water clarity, depth range, and preferred habitat of *Panulirus* species. Data mainly from Holthuis (1991), and personal observation. Depth: very shallow, 1-10 m; shallow, 1-20 m; moderate, 1-40 m; and deep, 1 → 40+ m

	Distribution	Local larval transport system	Depth range	Adult habitat
CLEAR WATER FIRST MAJOR LINEAGE				
<i>argus argus</i> (Latreille)	Caribbean	Caribbean Current, seasonal gyres	Deep	Coral + reef
<i>argus westoni</i> Sarver et al.	North-east Brazil	Brazil Current	? Deep	Algal caves
<i>interruptus</i> (Randall)	California	California Current, gyres + eddies	Deep	Rock
<i>japonicus</i> (Von Siebold)	S. Japan	Kuroshio Current, northern subgyre	Deep	Mudstone
<i>marginatus</i> (Quoy & Gaimard)	Hawaiian Is.	N. Pacific subtropical counter-current	Deep	Reef
<i>brunneiflagellum</i> Sekiguchi & George	Ogasawara Is.	Kuroshio Current, southern subgyre	Deep	Volcanic rock
<i>cygnus</i> George	Western Australia	Leeuwin Current, mesoscale eddies	Deep	Limestone reef
<i>pascuensis</i> Reed	Easter Is.	Island wake eddies	Deep	Volcanic rock
<i>longipes longipes</i> (A. Milne-Edwards)	Indian Ocean	Widespread	Shallow	Coral + rock
<i>longipes bispinosus</i> Borradaile	West Pacific	Widespread	Moderate	Coral + rock
<i>femoristriga</i> Von Martens	West Pacific	Widespread	Shallow	Coral + rock
<i>echinatus</i> Smith	Atlantic	Widespread	Shallow	Volcanic + rock
<i>penicillatus</i> (Olivier)	Indo-west Pacific	Widespread	Very shallow	Fore-reef
<i>penicillatus</i> 'Red', this paper	Galapagos	Central Pacific Counter-current	Very shallow	Volcanic rock
<i>guttatus</i> (Latreille)	Caribbean	?Caribbean Current	Very shallow	Fore-reef
<i>versicolor</i> * (Latreille)	Indo-west Pacific	Widespread	Very shallow	Rock + coral
TURBID WATER SECOND MAJOR LINEAGE				
<i>regius</i> De Brito Capello	West Africa	Canary Current	Moderate	Rock + silt
<i>polyphagus</i> (Herbst)	South East Asia	Bay of Bengal, seasonal gyre	Moderate	Rock + mud
<i>laevicauda</i> (Latreille)	N.E. Brazil	? Brazil Current	Moderate	Rock + fine shell
<i>gracilis</i> Streets	Panama Basin	Panama Basin eddies	Moderate	Patch reef + sand
<i>inflatus</i> (Bouvier)	Gulf of California	Gulf gyres + N. Equatorial Counter-current	Moderate	Rock + sand

TABLE I
(Continued)

	Distribution	Local larval transport system	Depth range	Adult habitat
<i>stimpsoni</i> Holthuis	South China Sea	Nainan, Taiwan + Kuroshio Currents	Moderate	Rock + silt
<i>homarus homarus</i> (Linnaeus)	Indo-west Pacific	Widespread	Shallow	Rock + sand
<i>h. megasculpta</i> Pesta	Arabian Sea	Arabian gyre + seasonal upwelling	Shallow	Rock + sand
<i>h. rubellus</i> Berry	S.W. Indian Ocean	Agulhas Current + coastal upwelling	Shallow	Rock + sand
<i>h.</i> 'Brown', this paper	Marquesas Islands	?Marquesas circulation	Shallow	Rock + sand
<i>ornatus</i> (Fabricius)	Gulf of Papua	Hiro Current + Coral Sea gyre	Shallow	Rock + silt

* *P. versicolor* prefers clear water, but on genetic and morphological evidence, belongs in the second major lineage.

systems, and ecology, arranged in probable order of emergence within each of the two major lineages.

Support for the distinctness of those species or subspecies comes from traditional morphological and more recently from genetic comparisons by Ptacek et al. (2001), Patek & Oakley (2003), and Ravago & Juinio-Menez (2003). Sarver et al. (1998) confirmed that the previously recognized differences in colour pattern between the Caribbean and Brazilian *P. argus* are supported by their genetic analyses and warranted subspecific recognition. George (1997) has presented the taxonomy of the Indo-west Pacific species of *Panulirus*. Three pertinent taxonomic updates have been published: by Chan & Ng (2001), and by Ravago & Juinio-Menez (2003) who correctly re-identified *P. longipes* '*shirahige*' (in George, 1997) as *P. longipes bispinosus* Borradaile, 1899, and by Sekiguchi & George (2005) who formally described *Panulirus* '*aka*' (in George, 1997) from the Ogasawara Islands as *P. brunneiflagellum*. For the remaining species of *Panulirus* that live in the Atlantic and the east Pacific, the taxonomic status of most species is quite clear (Holthuis, 1991; Sarver et al., 1998). These taxa are *P. argus argus*, *P. a. westoni* Sarver, Silberman & Walsh, 1998, *P. interruptus* (Randall, 1840) (although three sub-populations exist, Perez et al., 2001), *P. pascuensis* Reed, 1954, *P. guttatus* (Latreille, 1804), *P. regius* De Brito Capello, 1864, *P. laevicauda* (Latreille, 1804), *P. gracilis* Streets, 1871, and *P. inflatus* Bouvier, 1895. Each taxon inhabits a separate biogeographic zone similar to those recognized by Eckman (1967) and Luning (1990).

However, the precise taxonomic status of three widespread taxa, *P. penicillatus* (Olivier, 1791) and *P. h. homarus* (Linnaeus, 1758) in the Indo-Pacific and *P. echinatus* Smith, 1869 in the Atlantic, is uncertain because of their variations in morphology and colour.

In regard to *P. penicillatus*, McWilliam (1995) pointed out that in the puerulus phase, morphological variations in the posterolateral sternal spines have been described from Hawaii, Japan, and the Marquesas Islands. Adult *P. penicillatus* also show variations in background colours that include yellow-green through brown-green to blue-black or dark reddish-brown (Holthuis, 1991). The dark reddish-brown form is regionally restricted to the east Pacific, at the Galapagos, Clipperton, Socorro, and Revillagigedo Islands where it is known as the 'Red lobster' (Holthuis & Loesch, 1967; Holthuis, 1991). It probably represents an endemic in that Panama Basin region and it is extremely rare on the Mexican coast (Flores-Campaña & Pérez-González, 1991). McWilliam (1995) also noted variations in the degree of exopod reduction of the second maxilliped in the puerulus of *P. h. homarus*, described from the Indian Ocean, New Caledonia, and the Marquesas Islands. There is also colour distinction for the adults of *P. h. homarus* in the Marquesas where brown specimens have been reported and figured, in contrast to the olive-green in New Caledonia (George, 1972).

Such differences in morphology and colour warrant further genetic comparisons to discover whether sub-populations of *P. penicillatus* occur within its broad Indo-west Pacific range, and particularly whether an endemic sub-population of *P. homarus* exists at the Marquesas. Fish and molluscs have a high degree of endemism in the Marquesas due to their isolation and local current systems (Springer, 1982).

If, as I suspect, regional taxonomic or genetic distinction can be illustrated for sub-populations of *P. penicillatus* in the east Pacific, then Eckman's (1967) 'East Pacific Barrier' would represent a major distance barrier that maintains the isolation of those sub-populations. Until the above suspicions have been resolved, the east Pacific Red lobster will be referred to here as *P. penicillatus* 'Red' and the brown lobster at Marquesas will be referred to as *P. homarus* 'Brown'.

Panulirus echinatus is the third widespread and variable equatorial species. Vianna (1986) in his studies along the Brazilian coast and at some South Atlantic oceanic islands, recognized 'small spotted' and 'large spotted' *P. echinatus*. There were also 'intermediates' but he believed that there was insufficient evidence to justify subspecific separation. Preliminary genetic analyses of two samples of each of the two categories of *P. echinatus* from Brazil have been made by S. K. Sarver (pers. comm.) who wrote 'I think more specimens would need to be analysed to make any strong statements confirming recognition of the two forms. There is no evidence of differences between the two forms in my limited data'. Further

geographical studies on the morphology, colour, and genetics of *P. echinatus* throughout its range will be required to refine and expand our knowledge of the taxonomy of this variable species.

Affinities of *Panulirus* species

The value of the original four species groupings presented by George & Main (1967) has been critically reassessed by the phylogenetic studies based on phyllosoma and puerulus morphology by Baisre (1994) and McWilliam (1995), and the genetic studies by Ptacek et al. (2001) and Ravago & Juinio-Menez (2003). All authors found the four groupings useful but concluded that two clear divisions, rather than four, better represented the larval, puerulus, and genetic findings. Ptacek et al. (2001) stated “The first major lineage included all species of the morphologically based Groups I and II; the second lineage contained all species included in Groups III and IV (George & Main, 1967)” ... and ... “the observed pattern of sequence divergence is consistent with previous hypotheses that have suggested that species in Groups I and II represent an early radiation of *Panulirus*”... “but does not support the monophyly of any of the individual Groups I-IV”.

In her comparative assessments of larval and puerulus morphology, McWilliam (1995) modified the placements of some species between Groups I and II and between Groups III and IV but importantly not between Groups II and III, thus supporting the recognition of the two major lineages. The clades recognized in this paper are listed in table II, which sets out the species relationships as proposed by George & Main (1967), by McWilliam (1995), as well as some conclusions about the clade composition by Ptacek et al. (2001). McWilliam’s (1995) deliberations that led her to shift a particular species from one group to another are particularly noteworthy since they are based on the relative specialization of each species and therefore provide valuable assessments of the probable order of emergence of each species within a clade.

First major lineage

McWilliam (1995) removed the ‘intermediate’ *Panulirus interruptus* from Group I, transferring it to the second Group and she also transferred the probable ‘? *P. echinatus*’ from Group II to Group I. Her new second Group thus consisted of *P. interruptus* (2A), *P. guttatus* (2B) and *P. penicillatus* (2), none of which she believed is related to another. She regarded these as separate derivatives from Group 1 since each exhibits one or two larval modifications that become common morphological features in the more specialized species in her Groups 3 and 4. Ptacek et al. (2001) concluded that the north-east Pacific *P. interruptus* split off

TABLE II

Clades within the genus *Panulirus* as determined by morphological and genetic studies. The two major lineages are those identified after Ptacek et al. (2001)

Clades	Species	George & Main (1967) adults	McWilliam (1995) larvae	McWilliam (1995) puerulus	Ptacek et al. (2001) genetics
FIRST MAJOR LINEAGE					
<i>argus</i>	<i>argus</i> spp.	I	1	P1	<i>argus</i> -like ancestral species
stand-alone	<i>interruptus</i>	I	2A	P2A	1st split from <i>argus</i> ancestor
<i>japonicus</i>	<i>japonicus</i>	I	1	P1	sister to <i>longipes</i> clade
ditto	<i>marginatus</i>	I	1	P1	part of <i>japonicus</i> clade
ditto	<i>brunneiflagellum</i>	no data	no data	no data	no data
ditto	<i>cygnus</i>	I	1	P1	part of <i>japonicus</i> clade
ditto	<i>pascuensis</i>	I	no data	no data	part of <i>japonicus</i> clade
<i>longipes</i>	<i>longipes</i> spp.	I	1	P1	1st species of <i>longipes</i> clade
ditto	<i>femoristriga</i>	I	no data	no data	no data
<i>echinatus</i>	<i>echinatus</i>	II	1	P1	1st species of <i>echinatus</i> clade
ditto	<i>penicillatus</i>	II	2	P2	sister to <i>echinatus</i>
stand-alone	<i>guttatus</i>	II	2B	P2B*	not close to <i>echinatus</i>
SECOND MAJOR LINEAGE					
<i>regius</i>	<i>regius</i>	III	3	P3	1 st species of <i>regius</i> clade
ditto	<i>polyphagus</i>	III	4	P3	1 st split from <i>regius</i> clade
ditto	<i>laevicauda</i>	III	4	P3	part of <i>regius</i> clade
<i>gracilis</i>	<i>gracilis</i>	III	3	P3	no comment
ditto	<i>inflatus</i>	III	3	P3	no comment
stand-alone	<i>stimpsoni</i>	IV	4	no data	no comment
<i>homarus</i>	<i>homarus</i> spp.	IV	4	P4	the subspecies are sister taxa
stand-alone	<i>ornatus</i>	IV	4	P4	no comment
stand-alone	<i>versicolor</i>	IV	4	P4	no comment

* See also Briones & McWilliam (1997).

early from the western Atlantic *P. argus*-like ancestor and this was followed by the separation of the Pacific *P. japonicus* (Von Siebold, 1824) clade. Ptacek et al. (2001) also found that *P. echinatus* and *P. penicillatus* were well-supported sister taxa. If McWilliam’s (1995) identifications of the larvae and puerulus of *P. echinatus* are correct (and I see no reason why that conclusion is suspect), her groupings support the affinity of those latter two species. McWilliam (1995) suggested that *P. echinatus* is more likely to represent the ancestral species of a

P. echinatus clade since *P. penicillatus* has the specialized broadening of the larval cephalic shield and the shortening of the puerulus exopod of the third maxilliped. In the adult, the presence of double spines on the front of the antennal plate of *P. penicillatus* is also a unique specialization. Both McWilliam (1995) and Ptacek et al. (2001) include *P. guttatus* in Group II (2) but neither presents any suggestions for the provenance of *P. guttatus*, other than as a derivative from Group I. Ptacek et al. (2001) do not believe that *P. guttatus* shares a recent common ancestor with the *P. echinatus* clade.

In this paper, the recognized clades in the first major lineage (table II) are (a) the *P. argus* clade consisting of the two west Atlantic subspecies of *P. argus* (*P. a. argus* and *P. a. westoni*), (b) the Indo-west Pacific *P. japonicus* clade consisting of *P. japonicus*, *P. brunneiflagellum*, *P. cygnus* George, 1962, *P. marginatus* (Quoy & Gaimard, 1825), and *P. pascuensis* (c) the Indo-west Pacific *P. longipes* (A. Milne-Edwards, 1868) clade consisting of the two subspecies of *P. longipes* and *P. femoristriga* Von Martens, 1872, and (d) the *P. echinatus* clade consisting of the Atlantic *P. echinatus* and the Indo-Pacific *P. penicillatus*. The east Pacific *P. interruptus* and the Caribbean *P. guttatus* are regarded as 'stand-alone' species, probably as a result of early divergence (Ptacek et al., 2001).

Second major lineage

Based on puerulus stage comparisons within her Group 3, McWilliam (1995) gives good support for the grouping of *P. regius*, *P. polyphagus* (Herbst, 1793), and *P. laevicauda*. However, based on phyllosoma comparisons, she transferred *P. polyphagus* and *P. laevicauda* to her Group 4 because, unlike *P. regius*, which has no dorsal coxal spines and a narrower cephalic shield, they possess dorsal coxal spines and have a wide cephalic shield. These specializations suggest to me that *P. polyphagus* and *P. laevicauda* probably split off from a *P. regius*-like ancestor. Ptacek et al. (2001) also concluded that within this clade, *P. polyphagus* was the first species to split off, leading to my conclusion that *P. laevicauda* was probably the last to evolve in the *P. regius* clade.

Within the *P. gracilis* clade, the morphological differences between the very similar *P. gracilis* and *P. inflatus* have been clarified for adults by Holthuis & Villalobos (1962), and for late stage phyllosomata by Johnson & Knight (1966). Those two species were also considered to be closely related by McWilliam (1995) and Ptacek et al. (2001) but there are no comments in either of those texts that suggest the origin of that clade. All these authors place *P. gracilis* and *P. inflatus* in Group III (or 3), alongside the *P. regius* clade. Note that in George & Main (1967), their table III contains a publication misprint in placing *P. gracilis* and *P. inflatus* in Group IV. Ravago & Juinio-Menez (2003) grouped *P. homarus*, *P. ornatus* and

TABLE III
Origins and radiations of *Panulirus*

TIME A	TIME B	TIME C	PRESENT
FIRST MAJOR LINEAGE			
<i>argus</i> ancestor in central Tethys Seaway	<i>argus</i> extends from Brazil to Caribbean	eastward-flowing Amazon splits <i>argus</i>	<i>a. argus</i> <i>a. westoni</i>
<i>interruptus</i> in east Pacific	adapts to variable California Current	3 sub-populations adapt to fault movements	<i>interruptus</i> 3 sub-poplns.
<i>japonicus</i> ancestor in west Pacific	<i>japonicus</i> entrained in Kurschio subgyre	continued selection by shifts in subgyre	<i>japonicus</i>
	<i>brunneiflagellum</i> occupies emerging Ogasawara Is.	gears to southern Kuroshio subgyre	<i>brunneiflagellum</i>
	<i>marginatus</i> occupies emerging chain of Hawaiian Islands	gears to local circulation around Hawaiian Islands	<i>marginatus</i>
	<i>cygnus</i> adapts to Proto-Leeuwin Current	adapts to further Leeuwin Current development	<i>cygnus</i>
	<i>pascuensis</i> occupies emerging Easter Islands	gears to trapped island wake eddies	<i>pascuensis</i>
<i>longipes</i> ancestor in Indo-west Pacific	subspecies split initiated by the Pacific-Indian throughway closing	split almost physically complete	<i>l. longipes</i> <i>l. bispinosus</i>
	<i>femoristriga</i> origin unknown	adapts to equatorial west Pacific	<i>femoristriga</i>
<i>echinatus</i> ancestor in central Tethys Seaway	<i>echinatus</i> geared to wide-spread Atlantic circulation	shallow L spot form and deeper S form evolve	<i>echinatus</i> L <i>echinatus</i> S
<i>penicillatus</i> splits from <i>echinatus</i> stock into east Pacific	<i>penicillatus</i> stock expands into Pacific and Indian Oceans	<i>penicillatus</i> variants develop <i>penicillatus</i> 'Red' isolates in Galapagos region	<i>penicillatus</i> <i>penicillatus</i> 'Red'
	<i>guttatus</i> origin unknown	geared to coral fore-reef habitat	<i>guttatus</i>
SECOND MAJOR LINEAGE			
<i>regius</i> ancestor in broad Tethys Seaway	<i>regius</i> ancestor in central Tethys	<i>regius</i> isolates near Congo and Niger out-flows	<i>regius</i>
	<i>polyphagus</i> cut off in south east Asia region	separation re-enforced by Himalayan run-off	<i>polyphagus</i>
	<i>laevicauda</i> splits from <i>regius</i> stock	establishes near Amazon and Orinoco run-off	<i>laevicauda</i>
	<i>gracilis</i> ancestor splits from <i>regius</i> stock	adapts to new Panama Basin eddies	<i>gracilis</i>

TABLE III
(Continued)

TIME A	TIME B	TIME C	PRESENT
		<i>inflatus</i> splits from <i>gracilis</i> into Gulf of California	<i>inflatus</i>
	<i>ornatus</i> origin unknown	geared to Fly River/New Guinea mountains runoff	<i>ornatus</i>
	<i>homarus</i> origin unknown	'microsculpta' form gives <i>h. homarus</i> and <i>homarus</i> 'Brown'	<i>h. homarus</i> <i>h. 'Brown'</i>
		'megasculpta' form gives <i>h. megasculpta</i> and <i>h. rubellus</i>	<i>h. megasculpta</i> <i>h. rubellus</i>
	<i>stimpsoni</i> origin unknown	geared to Pearl River/Himalayan run-off	<i>stimpsoni</i>
	<i>versicolor</i> origin unknown	adapts to Indo-west Pacific	<i>versicolor</i>

Time A — Broad equatorial Tethys, but narrowing as Africa approaches Europe; India collides with Asia to initiate Himalayan mountains and give some run-off; broad, warm, temperature zones; current systems, e.g., Gulf Stream, California Current and Kuroshio present but ill defined; probable period: Late Oligocene-early Miocene (25-18 Ma).

Time B — African Plate hits Europe to make first barrier across Tethys; further slow uplifts of Himalayan Plateau; Australian Plate approaches Asian Plate; intensification of temperature zones, current systems and upwelling regions; probable period: Mid Miocene (18-8 Ma).

Time C — Isthmus of Panama forms final barrier across Tethys; the Andes rise to reverse Amazon flow into Atlantic; Gulf of California opens; intensified rise of Himalayan Plateau and New Guinea mountains; cooler conditions, stronger current systems and upwelling. El Niño established; probable period: late Miocene/Pliocene (8-2 Ma).

P. versicolor (Latreille, 1804) in a separate clade, but the topology of the species within the clade was not consistently resolved.

In this paper, the recognized clades in the second major lineage (table II) are: (a) the *P. regius* clade consisting of the West African *P. regius*, the mainly north Indian Ocean *P. polyphagus* and the western Atlantic *P. laevicauda* (b) the *P. gracilis* clade consisting of the central east Pacific *P. gracilis* and the Gulf of California *P. inflatus*; and (c) the *P. homarus* clade consisting of the three currently recognized subspecies of *P. homarus* (*P. homarus homarus* *P. h. megasculpta* Pesta, 1915, and *P. h. rubellus* Berry, 1974) as well as a fourth, unnamed subspecies, *P. homarus* 'Brown' from the Marquesas. The South China Sea *P. stimpsoni* Holthuis, 1963, the mainly Gulf of Papua *P. ornatus* (Fabricius, 1798), and the Indo-west Pacific *P. versicolor* seem to be stand-alone species of unknown ancestry.

Habitats, lineages, and clades of *Panulirus*

In the first major lineage (table I), the typical coastal habitats of all the adult lobsters are rocky or coral substrates, washed by clear, non-turbid 'oceanic' water. They live in subtropical and tropical regions along unsheltered coasts of continents or volcanic islands without significant riverine inputs, and therefore the coastal waters have undiluted salinity values at or above average oceanic levels. The local transport systems for larvae are usually large-scale oceanic current systems or meso-scale gyres.

By contrast, most species of the second major lineage live in the tropical monsoon regions where the coastal rocky crevice habitats of some species are often washed by turbid waters of low salinity, laden with fine terrigenous sediments flowing from small or large rivers. Other species are subjected to a different kind of turbidity because they live on open beaches where coarser, sandy sediments are frequently stirred up by the surf. The volume of river run-off and the severity of the surf may vary with monsoons or hurricanes so that the lobsters are tolerant of, but not necessarily dependent upon, turbid or low-salinity conditions. The local oceanic systems for the transport of the larvae of these species are usually moderate sized or small-scale circulations (<50 km diameter) such as coastal upwelling and seasonal gyres.

The *Panulirus* ancestral stock

Both *Panulirus argus* and *P. longipes* have been regarded as the best candidates to represent the most primitive ancestor of *Panulirus* since both species have very wide environmental tolerances in both the larval and the adult condition (George & Main, 1967). Those authors chose the Indo-west Pacific *P. longipes* to best represent the ancestral species only "because it is best known to us". However, all three sets of genetic data obtained by Ptacek et al. (2001) favoured a *P. argus*-like ancestor. If this ancestor lived in the Atlantic then *P. argus* represents an Atlantic relict species. George & Main (1967), McWilliam (1995), and Ptacek et al. (2001) all agree that the first major lineage (clear water inhabitants) contains the species that evolved earlier than those species in the second major lineage (turbid water inhabitants). Clear oceanic-water habitats and coastal turbid habitats caused either by river run-off or surf, have probably been equally common ecosystems throughout the Mesozoic and Tertiary but it is not possible to conjecture how and under what conditions the *Panulirus* ancestors were split between the two major habitat types. It is postulated here that once the clear and the turbid water habitats had been established, each of the two "ecological ancestors" diverged as each of the ancestral habitats became further subdivided by subsequent physical and environmental changes.

Origin and emergence times of *Panulirus*

To select only one species that represents **the** ancestral species of *Panulirus* may be premature at this time especially since Ptacek et al. (2001) state that "Relationships within major lineages are not well resolved" . . . "the molecular phylogeny does not support the monophyly of any of the individual Groups I-IV". It seems probable, however, that the establishment and subsequent subdivision of the first major lineage occurred very early in the evolution of *Panulirus* (table III). It is proposed that the ancestor of *P. argus* was centred in the Atlantic sector of the Tethys Seaway, that of *P. interruptus* centred in the east Pacific, that of *P. japonicus* centred in the west Pacific, and that of *P. longipes* centred in the Indo-west Pacific. *Panulirus echinatus* was initially widespread across the open Tethys Ocean and ultimately provided *P. penicillatus* as an isolated daughter species in the Pacific.

In the absence of any fossils of the genus *Panulirus*, its actual emergence time and subsequent history of radiation must be speculative. George & Main (1967) had suggested a post-Pliocene emergence. However, allozyme comparisons and cautious estimates of emergence times by Chan & Chu (1996) and by J. Macaranas (pers. comm. in George, 1997), prompted George (1997) to propose that a Miocene or earlier commencement was required to account for the radiation of *Panulirus* in the Indo-west Pacific region. He (George, 1997) also believed the most recently evolved species, *P. versicolor*, was already established in the Pliocene.

George (1997) considered *P. polyphagus* to have been established by at least 7 Ma (late Miocene), or perhaps earlier, after the Himalayas had risen during the Miocene (11-23 Ma). Ptacek et al. (2001) regarded *P. polyphagus* as the first species to split from *P. regius* (see table II) and the timing of the split provides a crucial key to the possible emergence time of the second major lineage (table III). It is proposed that during the Miocene (fig. 1), the turbid water *P. regius* ancestor lived all the way across the Tethys Seaway from the Atlantic and into the Indian Ocean and that the Levant closure isolated *P. polyphagus* in the Indian Ocean in the late Miocene.

Radiation of the first major lineage

Panulirus argus

The ancestral stock of *P. argus* is thought to have had a wide environmental tolerance (George & Main, 1967) and probably occupied an uninterrupted distribution in the clear water tropical and subtropical regions of the western Atlantic at least from Florida to Brazil. The significant barrier to gene flow that isolates the separate clear water habitats of the two subspecies of *P. argus* today, is the 1100 km coastal region of low salinity, high turbidity, and heavy sediment deposits

near the mouths of the Orinoco, Amazon, Tocantins, and Parnaíba rivers (Sarver et al., 1998).

The collision of the oceanic Nazca Plate with South America and the resulting Andean orogeny had profound effects on the riverine inputs into both the Pacific and the Atlantic Ocean in the Central American region. According to Sterling (1984), before the mid-Miocene Andean uplift 15 Ma, the Amazon Basin, bordered by the Guyana Highlands shield to the north and the Brazilian Highlands shield to the south, emptied its freshwater and sediment load northward into the Caribbean region and further westward into the Pacific. When the Andes first arose to block the north and west outflows, the Amazon Basin formed a huge land-locked lake. It was not until late Miocene, about 10 Ma, that the run-off from the lake flowed eastward as the mighty Amazon River to enter the Atlantic and thus separate the two subspecies.

Panulirus interruptus

P. interruptus is represented by three subpopulations (Perez et al., 2001) and has many unique morphological features that, when combined, make it very difficult to assess its phylogeny (McWilliam, 1995). George & Main (1967) and Pollock (1992) proposed an Indo-west Pacific *P. longipes* ancestor for *P. interruptus*. However, it is now believed that *P. interruptus* was the first species to split off from the *P. argus* ancestor (Ptacek et al., 2001), migrating north-westwards into the subtropical East Pacific, near Baja California.

In southern California and Baja California, the coastal waters are currently under the influence of the unstable southern sector of the south-flowing California Current System that is constantly developing meanders and isolated eddies (Pringle, 1986). Because of that instability, it is not surprising that Johnson's (1960, 1971) attempts to correlate larval recruitment paths with the series of semipermanent eddies along the coast, failed to obtain a clear, simple picture.

The present and the past coastal hydrodynamics along the Californian coast, as well as the genetics of *P. interruptus*, are indeed complex. Perez et al. (2001) revealed that only the central sub-population of *P. interruptus* should be regarded as a panmictic population, and they emphasize that discrete physical oceanographic phenomena occur in each of the three geographic regions that are occupied by the separate genetic sub-populations. The discrete environmental phenomena are the Southern California Eddy in the north, the summer upwelling and winter cyclonic eddies in the central region, and the interactions between the California Current and the Equatorial Counter-current in the south.

According to Vega (2003), the timing and duration of mating, oviposition, incubation, and hatching in *P. interruptus* in the central region of Baja California is strongly dependent on El Niño-La Niña seasonal variations. This "relationship

between reproduction and the environment suggests that the reproductive pattern and larval recruitment are coupled to the dominant physical features of the California Current" in that region.

An analogue of the California Current was probably developed in the Cretaceous, and by the Miocene it was a well-established phenomenon (Kennett, 1982). However, the history of the California Current since then has probably been as torrid and unstable as the adjacent, north-sliding landmass that has shifted Pt. Conception 300 km northward of its Pliocene (11 Ma) position (Newman, 1979). As bits of Mexico were rafted northward alongside the San Andreas fault, the tectonically active sea-floor would have been modified by landslides, island emergences, and canyon formations. These submarine distortions would have broken up the coastal geomorphology as well as the path of the California Current, thus forcing the associated faunas to rapidly adjust to any new swirls, eddies, local currents, and offshore or onshore deviations. According to Flannery (2001), the geographic "trumpet shape" of North America has been instrumental in amplifying past climatic changes and has resulted in heightened seasonality and climate instability. Offshore winds that blow in summer through gaps in the mountains cause local up-welling and eddy formation, and the many geological and erosional changes in the configuration of the Rocky and Sierra Nevada mountains, would have shifted the upwelling areas to other coastal locations. Thus, during the evolution of the southern California marine fauna, each species has had to adjust to frequent and substantial shifts in the position, strength, and seasonality of the California Current System.

In addition to *P. interruptus*, many Californian species, such as benthic barnacles (Newman, 1979) and eight species of fish (Hewitt, 1981; Waples & Rosenblatt, 1987), have been shown to have sub-populations in close geographic proximity. Hewitt (1981) concluded that limited communication between the semipermanent eddies of the region is a significant speciation mechanism that would be enhanced by precise spawning times coinciding with the seasonal formation of each eddy system.

A north-south temporal cline in reproduction of *P. interruptus* has been reported by Perez et al. (2001) and of the two explanations proposed by them for the evolution of the three separate sub-populations of *P. interruptus*, I favour their first concept that focuses on the separate physical oceanographic phenomena within the California Current System. Through natural selection, that system has shaped and reshaped spawning times, larval behaviour, and puerulus longevity to produce the successful evolution of the three population units. Their alternate proposal of "sweepstake-chance matching" to account for the observed "chaotic genetic patchiness" may provide an explanation for small-scale sub-populations (Larson

& Julian, 1999) but whether it applies to long-lived animals with long pelagic larval lives, like spiny lobsters, remains to be determined.

The palaeoecological history of the subtropical *P. interruptus* has been the most turbulent and unstable of all the *Panulirus* species, perhaps explaining why it is such an enigma in its morphological and genetic characteristics (McWilliam, 1995; Ptacek et al., 2001).

The *Panulirus japonicus* and *P. longipes* clades

The following discussion of the probable evolution and radiation of the eight taxa from the *P. japonicus* and *P. longipes* ancestral stocks is facilitated by the recognition of three separate biogeographic zones in the Indo-west Pacific. These are: (a) the north-west Pacific Ocean subtropical zone containing *P. japonicus*, *P. brunneiflagellum*, *P. marginatus*, and *P. longipes bispinosus* (the northern sub-population, Japan to Philippines, see below); (b) the South Pacific subtropical zone containing *P. longipes bispinosus* (the southern sub-population, Australia to Polynesia), *P. cygnus*, and *P. pascuensis*; and (c) the equatorial tropical zone that contains *P. longipes longipes* and *P. femoristriga*.

The *Panulirus japonicus* clade

This clade is regarded as a sister clade of the *P. longipes* clade (Ptacek et al., 2001) and is here considered as an early separation (Ravago & Juinio-Menez, 2003) that became isolated in the west and central Pacific sector of the Tethys Sea following the African-Asian collision in mid-Miocene (Veron, 1995).

Panulirus japonicus

An analogue of the Kuroshio Current developed east of Japan as long ago as the Cretaceous (Kennett, 1982) and *P. japonicus* is seen as an isolate adapting to a larval transport mechanism that was provided by this long-lived current system. In the Japanese region, major volcanic pulses peaked in the mid-Miocene, resulting in lower water temperatures due to upwelling of cold water in late Miocene (Kennett, 1982). The Kuroshio Current also intensified, spinning off local gyres and forming the discrete Kuroshio Counter-current subgyre (George, 1997; Sekiguchi, 1997; Inoue & Sekiguchi, 2001). Evolution and modification of larval behaviour would have been essential to maintain recruitment success via the clockwise and anti-clockwise circulations south of Japan and the developing sub-gyres of the Kuroshio Counter-current system.

Panulirus marginatus

The Hawaiian endemic *P. marginatus* is probably a central Pacific relict species from the formerly widespread subtropical *P. japonicus* ancestor. The volcanic Hawaiian Island chain has been forming over the Hawaiian hotspot and eroding to seamounts for about 60 Ma (Springer, 1982) and since the Oligocene, the general North Pacific circulation has intensified (Kennett, 1982). Perhaps in the late Miocene, *P. marginatus* evolved the larval retention strategies that were necessary to achieve satisfactory recruitment pathways within the boundaries of the Hawaiian endemic oceanic system (Polovina et al., 1995).

Panulirus brunneiflagellum

This species probably split off from the Japanese *P. japonicus* ancestor following further west Pacific volcanic action along the Iwo Jima Ridge, south of Japan (Sekiguchi & George, 2005). The Ogasawara Island chain is the youngest (2 Ma) extension of the West Mariana fore-arc that had formed in the early Miocene about 20 Ma (Springer, 1982). The emergence of these oceanic island chains and ridges would have provided a new offshore habitat for *P. brunneiflagellum* that was supported by a new and reliable recruitment pathway for its larvae in a separate subgyre to the south of, and independent of, the Kuroshio Counter-current (Sekiguchi, 1997). Whether *P. brunneiflagellum* split off at the same time as the West Mariana fore-arc formed (20 Ma), or at a later time when the Ogasawara Chain extended (2 Ma), remains to be resolved (Sekiguchi & George, 2005).

Panulirus cygnus

The larval transport and recruitment pathways of *P. cygnus* are geared to an adjacent current system, the Leeuwin Current (Griffin et al., 2001). *P. cygnus* is the separated daughter species of the *P. japonicus* ancestor and the larvae have adapted to the mesoscale eddies off the west coast of Western Australia, as the climate warmed (Pollock, 1992; George, 1997). Griffin et al. (2001) modelled the variables of the Leeuwin Current and the El Niño Southern Oscillation (ENSO) that impact on the annual recruitment of *P. cygnus*, and these variables highlight the significance of local oceanic transport systems in the evolutionary processes of *Panulirus*. According to Shafik (1990), nanoplankton analyses show that a strong Proto-Leeuwin Current flowed down the west coast of Australia and into the Great Australian Bight several times during the Eocene and middle to late Oligocene. Since Pliocene times, he (Shafik, 1990) also reported that variations in the strength of the Leeuwin Current due to El Niño-La Niña oscillations have probably operated to some degree as far as the eastern Great Australian Bight. These events have been part of the natural selection mechanisms that have resulted in the successful and flexible biological and genetic strategies evolved by *P. cygnus*.

Panulirus pascuensis

P. pascuensis is part of a highly endemic Easter Island fauna (Springer, 1982) that inhabited the succession of shallow seamounts and islands that preceded the actual emergence of Easter Island itself. Pollock (1992) suggested that these emergences together with the Pliocene development of a southern Pacific palaeocirculatory cell effectively isolated *P. pascuensis* from the generalized *P. longipes* ancestor (= *P. l. bispinosus*, southern component: see above). Distribution records of *P. l. bispinosus* indicate that the Cook Islands in Polynesia are the most easterly present-day localities of the *P. longipes* stock. Perhaps the failure of their larvae to disperse eastward against the northwest current as depicted by Pollock (1992, his fig. 3) is the reason for its inability to occupy Easter Island today.

Springer (1982) discussed the vicariant geological events that led to the biotic isolation of the Easter Island fauna and stated "Until about 20 my ago, there was a common biota inhabiting the islands of south-east Oceania on both sides of the East Pacific Rise at about the latitude of Easter Island. The uniformity of this biota was maintained in the face of diverging oceanic lithospheric plates (seafloor spreading) and the subsidence and drowning of islands on each side of the East Pacific Rise by the periodic formation of new islands within biotic dispersal distance on each side of the Rise. Island formation on the Nazca Plate in the vicinity of Easter Island has been continuous, but about 20 my ago, island formation on the Pacific Plate adjacent to the same vicinity ceased. Westward movement of the Pacific Plate gradually increased the distance between Pacific Plate islands and islands of the Easter volcanic chain until interchange between these two groups of islands was no longer possible. The resulting isolation of the more western islands of the Easter volcanic chain allowed many species in these islands to diverge, accounting for the high degree of endemism that we recognise today".

The infrequent presence of *P. pascuensis* (cf. Holthuis, 1991) on the young (<2 Ma) volcanic Pitcairn Island is probably the result of the diminishing recruitment by larvae originating from Easter Island. The continuing movement of Pitcairn Island further away from Easter Island on the westward-moving Pacific Plate further reduces the likelihood of regular recruitment and indicates the final stages of "larval stretch and break" (George, 1997).

The Panulirus longipes clade

The following scenario is proposed for the radiation of the *P. longipes* clade. The *P. longipes* ancestor previously occupied a wide equatorial tropical Indo-Pacific region when surface water temperature gradients were broad and currents flowed freely between the Pacific and the Indian Ocean (Frakes & Kemp, 1973). As Australia moved northwards to narrow the through-way between the Pacific and the Indian Ocean in late Miocene (Audley-Charles, 1981), *P. l. longipes* gradually

became more isolated in the Indian Ocean, although it still remains today in the Philippines overlap area with *P. l. bispinosus*, where hybridization occurs (Juinio et al., 1991). Ravago & Juinio-Menez (2003) found that the two subspecies are genetically very closely related and probably diverged less than 1 Ma. Uthicke et al. (2004) similarly suggested that the separation of these two oceans and altered current patterns during the Pliocene had resulted in the allopatric speciation of two species of teat fish, *Holothuria whitmaei* Bell, 1887 and *H. nobilis* (Selenka, 1867).

The west Pacific *P. longipes bispinosus* lost some of its warm water tropical characteristics, retained more of the subtropical, deeper water characteristics and thus became divided into the northern hemisphere and southern hemisphere sub-populations as the surface water isotherm gradients steepened and the oceanic circulation systems intensified (Kennett, 1982).

The two sub-populations now have out-of-phase summer spawning times in these two subtropical regions. Oceanic circulation east of the South China Sea is centred to the south of the Kuroshio Countercurrent Subgyre and this circulation is the main transport system for the return of the puerulus of the northern sub-population of *P. l. bispinosus* (cf. Inoue & Sekiguchi, 2001). For the southern sub-population of *P. l. bispinosus*, the South Equatorial current is probably a dominant influence. Dennis et al. (2001) suggest that the larvae that are released in summer in the Great Barrier Reef, Queensland are probably reliant upon it for the successful recruitment to the eastern Coral Sea and adjacent regions. McWilliam & Phillips (1983) tracked the passage of wayward *P. l. bispinosus* larvae southward down the east coast of Australia from August 1979 to May 1980 in a warm-core eddy (Eddy J). They concluded that these late stage larvae had been transported too far south and were unable to return to their coast of origin.

The shallow water equatorial *P. femoristriga* stabilized in the central West Pacific tropic region (George, 1997) but its origin is not known.

The *Panulirus echinatus* clade

Panulirus echinatus

Ptacek et al. (2001) suggested an early split of *P. echinatus* from *P. argus*. I suggest an almost simultaneous secondary separation of the Indo-Pacific *P. penicillatus* from the Atlantic *P. echinatus* stock in the Miocene when separate off-shore oceanic current systems in both the Atlantic and Pacific were well established, much as they are today (Kennett, 1982). The Panama Seaway was open and the Amazon basin drained westward into the central American region (Sterling, 1984) probably supplying a significant turbid water barrier between the Atlantic *P. echinatus* and the separated Pacific *P. penicillatus*. The development of a further physical barrier between the two species finally completed the isolation when the Panama Seaway closed during the Pliocene.

In the Atlantic, the ocean current systems that are so vital for the widespread transport of larvae of *P. echinatus* intensified, and genetic exchange within the North and South Atlantic Basins were possibly altered, perhaps producing the 'large spot' and 'small spot' variants as described by Vianna (1986).

Panulirus penicillatus

The above model proposed for *P. echinatus* could also be proposed to explain morphological variations and possible sub-population isolation in *P. penicillatus* throughout the Indo-Pacific. Perhaps the exceptional dispersal qualities of the larvae of the *P. penicillatus* ancestor enabled widespread genetic exchange throughout the Pacific. At that time, the temperature zones were wide and the currents were weak, but the intensified circulation in the late Miocene would have localized larval transport around island chains such as the emerging Hawaiian Chain (Polovina et al., 1995) and the Galapagos Chain. For the isolated *P. penicillatus* "Red" on the Galapagos and adjacent islands, the eastward movements of the Galapagos Islands on the Nazca Plate continued to carry that sub-population further from the central Pacific populations of *P. penicillatus* and thus widen the oceanic East Pacific Barrier.

Panulirus guttatus

There are little genetic or morphological data to indicate the relationships of the stand-alone *P. guttatus*. It is an obligate reef-dweller and all the benthic phases of its life cycle have adapted to a narrow habitat in the very shallow spur and groove zone of the fore reef (Briones & McWilliam, 1997; Sharp et al., 1997).

The adult habitat of *P. guttatus* is very similar to that of the two species in the *P. echinatus* clade that also live in the very shallow, reef-front zone (table I) and for that reason, I suggest that *P. guttatus* probably evolved from the North Atlantic *P. echinatus* stock. Although Ptacek et al. (2001) deny that the *P. echinatus* clade shares a recent ancestry with *P. guttatus*, a split could have taken place in early Miocene, since Veron's (1995) study of the evolution of corals shows that in the Caribbean, such coral reef habitats had been available as far back as the Late Cretaceous (75 Ma). In response to the same prevailing Caribbean current systems, the larvae of *P. a. argus* and *P. guttatus* must have evolved significantly different behavioural characteristics in order to retain their individuality. Perhaps the larger sizes of the egg, phyllosoma, and puerulus of *P. guttatus* (cf. Pollock, 1997) are some of the morphological evolutionary adaptations that have been adopted, allowing its own distinctive life history to develop and the puerulus to detect and settle successfully in its rather narrow habitat.

Radiation of the second major lineage

The *Panulirus regius* clade

The major deltas or cones that provide the fine-grained sedimentary substrates for the three species that make up the *P. regius* clade are the Niger and Congo cones (*P. regius*), the Bengal and Indus cones (*P. polyphagus*), and the Amazon and Orinoco cones (*P. laevicauda*). These three species are usually found in low salinity at moderate depths and in sandy or muddy conditions (table I).

It is postulated that in the early Miocene, the original *P. regius* stock would have been living in the Tethys Sea along the riverine-influenced western and southern coasts of Europe, the western and northern coasts of North Africa (Kennett, 1982) and further eastwards into the Indus and Bengal regions of south east Asia (fig. 1). *Panulirus regius* survives today in the relict habitat along the West African coast that receives heavy Hinterland discharges from the ancient Niger and Congo Rivers.

As Africa moved northward to collide with Eurasia in the Arabian region in the late Miocene, the incipient *P. polyphagus* would have been split off physically from the *P. regius* stock and isolated in the Indian region. The initial Eocene collision of India with Asia had already occurred, uplifting the Himalayan mountains that provided the huge river discharges that have supplied and still supply today the immense Indus and Bengal rivers and their deltaic cones (George, 1997). This sediment covers rocky outcrops and forms the extensive mud-banks where *P. polyphagus* lives today (Bhatia, 1974).

Following the major Andean orogeny (see radiation of *P. argus*), the reversal of the Amazon flow into the Atlantic provided a new habitat for *P. laevicauda* on the northern Brazilian coast about 10 Ma (Sterling, 1984). Separation of *P. laevicauda* from the *P. regius* stock was enhanced as each species adapted to the two distinct depositional histories and asymmetrical transport currents of the south eastern and south western North Atlantic basins (Kennett, 1982). The subsequent physiological and behavioural responses by *P. laevicauda* to tolerate the Amazon's massive input in turbidity and decrease in salinity are not known, nor the larval behaviour or recruitment pathways. Perhaps *P. laevicauda* progressively migrated away from the excessive smothering effects of the heavy freshwater flow from the Amazon and sought marginal areas in northeast Brazil, which received only moderate amounts of silt covering its rock shelters.

The *Panulirus gracilis* clade

The ancestral source of the *P. gracilis* clade is not clearly indicated by previous authors but George & Main (1967), McWilliam (1995), and Ptacek et al. (2001) all place the two species together with the *P. regius* clade in Group III. I propose

that the *P. gracilis* ancestral stock diverged from the *P. regius* ancestral stock by adapting to a coarser-grained, sandier habitat compared with the finer-grained, muddy habitat of the *P. regius* stock.

Panulirus gracilis

Before the Miocene uplift of the Caribbean belt in Colombia and Venezuela and that of the Andes (Read & Watson, 1985), the Amazon River would have drained westward into the southern Caribbean and on into the Pacific. This drainage may have supported the initial occupation of the Central American region by the turbid-tolerant *P. gracilis* ancestor. After the Miocene orogenies of the Caribbean belt and the Andes, the establishment of *P. gracilis* would have been sustained by the many small rivers and the low surf along the central east Pacific coast. The dominant features controlling larval retention in the Mexico-Ecuador-Galapagos region are two major eddies in the Panama Basin, one circulating cyclonically around the Costa Rica Dome at about 9°N 89°W and the other is an anticyclonic eddy centred about 5°N 88°W (Johnson, 1971). That author believed that larvae carried well beyond the Galapagos Islands by the South Equatorial Current were considered lost. At what time these two major eddies originated and what factors controlled their persistence are unclear but the Pliocene closure of the Panama Seaway most likely assisted in their definition and hence in the ultimate speciation of *P. gracilis*.

The oldest, easternmost islands and sea-mounts of the Galapagos Chain emerged as a potential lobster habitat about 5-10 Ma. The successful migration of *P. gracilis* larvae from South America was made easier because the whole chain of the Galapagos Islands has been moving on the Nazca Plate and closer to Central America at the rate of 7 cm each year (Jackson, 1993).

Panulirus inflatus

P. inflatus is seen as a recent split from the *P. gracilis* stock into a new habitat in the Gulf of California when the peninsula of Baja California initially separated from mainland Mexico in the late Miocene (Alvarez-Borrego, 1983). About 4 Ma, the rate at which the opening of the Gulf opened increased, and its present opening rate is about 6 cm per year. Alvarez-Borrego (1983) and Aguirre-Gomez et al. (1999) detail the unique oceanographic environmental conditions in the Gulf of California. These features include: (a) a bottom topography of a steep-sided narrow, desert-bound gulf, subdivided into several isolated basins; (b) highly variable sea surface temperature (15-20°C in winter, 27-29°C in summer); (c) very high salinity due to a high evaporation rate (130 cm/yr); (d) a complex pattern of hydrological and seasonal currents; (e) a unique tidal regime of high spring tides of high velocity (Filloux, 1973); and (f) unique upwelling events, particularly between March and June.

Such conditions would have placed strong physiological and behaviour selection pressures on larvae, pueruli, and adults to adapt in only about 6 My since the opening of the Gulf. Nevertheless, only slight morphological, colour pattern, and genetic differences between *P. inflatus* and *P. gracilis* are apparent (Holthuis & Villalobos, 1962; McWilliam, 1995; Ptacek et al., 2001).

Concentrations of mid and late stage larvae of *P. inflatus* have been recorded by Johnson (1960, 1971) along the western coast of Baja California, closely paralleling its adult distribution and the author noted that the area is not affected by the southward flow of the California Current. From Johnson's (1971) chart 13, it appears that *P. inflatus* larvae are closely associated with the coastal section of the North Equatorial Countercurrent and he draws the general conclusion that recruitment "probably depends largely upon retention of larvae in the coastal areas by eddies or coastal countercurrents".

The coastal current system that now successfully transports and returns the larvae of *P. inflatus* has likely developed since the Pliocene closure of the Panama Seaway. The characteristics of that current played a vital part in the selection processes of *P. inflatus* as it evolved to occupy its new habitat in the Gulf of California and on adjacent coasts.

Panulirus stimpsoni

The larval cycle and the recruitment pathway of *P. stimpsoni* are not known but the response of larvae to the seasonal effects of the Asian monsoon cycle and the nearby Kuroshio, Hainan, and Taiwan currents are presumed to play a large part in its evolutionary success as the dominant spiny lobster species in the South China Sea. Its habitat and local oceanic circulatory processes were probably developing in the Miocene when run-off from the rising Himalayas discharged into the South China Sea and the seasonal monsoon system was well established (George, 1997).

The *Panulirus homarus* clade

Three regionally separate subspecies in the *P. homarus* clade inhabit the Indian Ocean and the western side of the Pacific, albeit with low percentages of overlap (Berry, 1974a; Hadkirasouliha, 1994). Discrete larval pools (Pollock, 1993) keep each subspecies more or less geographically separate. The fourth subspecies is a well-separated taxon in the east Pacific. Two subspecies, *P. h. megasculpta* and *P. h. rubellus* have large, well-developed scalloping along the abdominal transverse grooves (the '*megasculpta*' subclade) and the other two, *P. h. homarus* and *P. h. "Brown"* possess low scalloping along these grooves (the '*microsculpta*' subclade) (see table III).

Panulirus h. megasculpta

The northern Indian Ocean subspecies *P. h. megasculpta* has become isolated in the Arabian Sea and specifically geared to the conditions of the alternating NE-SW monsoon system that was initiated by the Himalayan orogeny. As the Himalayas rose higher, the monsoon system strengthened and caused strong upwelling in the Arabian Sea, a feature that has been present since the Miocene (Kennett, 1982; George, 1997).

Panulirus h. rubellus

Recruitment to the southern Indian Ocean subspecies, *P. h. rubellus*, of south eastern Africa is largely dependent on coastal reversing gyres that occur inshore of the Agulhas Current (Berry, 1974b). It has been proposed that the strengthening of the whole Agulhas system generated the distinctive nature of the inshore coastal system and has resulted in the geographic separation of *P. h. rubellus* from *P. h. megasculpta* (cf. George, 1997).

Panulirus h. homarus

The third subspecies, *P. h. homarus*, is widespread across the Indian Ocean and into the west Pacific. Nothing is known about its recruitment pathways over such vast distances. George (1997) estimated its emergent time to be mid Pliocene (ca. 3.4 Ma) and its ability to hybridize with *P. h. rubellus* and *P. h. megasculpta* (cf. Berry, 1974a; Hadkirasouliha, 1994) indicates the relatively recent separations of the whole subspecies complex.

Panulirus h. "Brown"

The fourth subspecies, *P. h. "Brown"*, is proposed here to be a Marquesas endemic because of its geographic isolation in the Marquesas Islands, which are situated 500 km from their nearest island neighbour, the Tuamotus. The Marquesas Islands are situated on a short, 350 km volcanic chain made up of recent islands in the south that emerged about 1.3 Ma and older islands in the north that emerged about 5.1-8.7 Ma (Springer, 1982). To explain the high degree of endemism that is found in Marquesan fish and molluscs, Springer (1982) proposed that "past local ocean currents (or eddies) that had permitted biotic dispersal from the Tuamotu region to the Marquesas were gradually modified (deflected) and eliminated as the Marquesas increased in linear extent". It is proposed here that the southern linear island extension in the Pliocene enhanced a strong, localized oceanographic circulatory system around the Marquesas to which the larvae of *P. h. "Brown"* have successfully adapted.

Panulirus ornatus

P. ornatus is a stand-alone species that is the dominant species in the Gulf of Papua region. Dennis et al. (2001), in their study of the larval and pueruli distributions of *P. ornatus* in the Coral Sea, state that phyllosoma larvae "are transported from the Gulf of Papua breeding grounds by the Hiri boundary current into the Coral Sea Gyre and then by surface onshore currents onto the Queensland coast and into Torres Strait".

These conclusions support George's (1997) hypothesis that the Gulf of Papua provides a stable regional environment together with an adjacent, specialized circulatory system into which *P. ornatus* evolved following the New Guinea tectonic uplift and the subsequent Fly River discharge in the late Miocene. *P. ornatus* has one of the shortest larval lives, of only 4-7 months (Skewes et al., 1997) and with its once-only breeding migration at 2+ years (Dennis et al., 2001) is the shortest-lived, fastest-growing species of *Panulirus*.

Panulirus versicolor

P. versicolor is believed to be the most recently evolved species of *Panulirus*, perhaps emerging in the Pliocene, about 2 Ma (George, 1997). Its genetic pattern and its morphology clearly place it in the second major lineage but unexpectedly, its habitat preferences for clear-water and coral reefs are those found in the species of the first major lineage. Unfortunately, the studies by McWilliam (1995) and Ptacek et al. (2001) throw little light on its possible ancestral history and therefore it is regarded here as a stand-alone species of unknown origin. More detailed investigations into its ancestors, its relatives, its adult and larval characteristics, as well as its ecological requirements are needed.

CONCLUSION

In the particular cases used in this review, the current diversity in the genus *Panulirus* may be explained through the interaction between geographical history and biological (i.e., taxonomic) units. The stability of, or the changes in, the circumstances governing the habitats of the spiny lobsters at each phase of their life cycles have set the scene for speciation events in the past, some of which have resulted in major splitting of lineages. Cladistic pattern analysis, taking into account morphological as well as molecular data in an integrated approach, should confirm, modify, or maybe even reject the correctness of the present conclusions.

ACKNOWLEDGEMENTS

I wish to thank John Lowe for his companionship and forceful discussions during our three-month tour in 1995 during which we visited the Smithsonian

Institution, the Florida Phylogeny Workshop, Puerto Morelos and Mazatlán in Mexico, Manta in Ecuador, the Galapagos Is., Juan Fernandez Is., Easter I., and Marquesas Is. Thanks are also due to Margaret Ptacek, Michael Childress, Shane Sarver, and Bill Herrnkind who all made such a success of the 1995 Phylogeny Workshop. Patricia Briones-Fourzán at Puerto Morelos and Raoul Pérez-Gonzales at Mazatlán kindly provided much-appreciated personal and institutional support during our visits there. During my visit to Chichi-Jima, Ogasawara in 1998, the personal guidance and assistance provided by Hideo Sekiguchi and the staff at the Ogasawara Fisheries Centre is appreciated. I dedicate this paper to Ms. Paulette McWilliam and to Professor Bert Main, whose scientific discussions during and after my University days repeatedly spurred my quest for a deeper understanding of spiny lobster biology, ecology, taxonomy, and evolution.

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